

Natalie Lui

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EDUCATION

University of California, Irvine (UCI)

Expected: June 2027

B.S. Computer Science, Minor in Informatics

- **Courses:** Design and Analysis of Algorithms (C++), Data Structures and Algorithms, Machine Learning & Data Mining, Next Generation Search Systems, Intro to Artificial Intelligence, Human Computer Interaction, User Interaction Software

SKILLS

Languages: JavaScript, Python, Java, C++, C#, SQL, HTML, CSS | **Tools and Frameworks:** Node.js, REST APIs, Git Version Control, GitHub | **Design & Product Tools:** Figma, Adobe Photoshop, Adobe Illustrator, Generative AI

UX Engineering Skills: Prototyping, Responsive UI Development, User Research, Usability Testing

TECHNICAL EXPERIENCE

Yuka Cafe

Dec. 2025 - Present

Software Engineer

- Developed a full-stack online ordering platform enabling customers to place orders and complete payments.
- Built frontend components using HTML, CSS, and JavaScript, ensuring responsive design across devices.
- Implemented backend services using Node.js and Square APIs for order processing, payment integration, and application state management.
- Collaborated with designers and stakeholders to translate product requirements into deployed features.
- Iterated on features and deployed updates using Git-based version control workflows.

UCI Multimedia Resource Center | Irvine, CA

Apr. 2024 - Present

Peer Technology Trainer

- Provided technical support for 100+ students weekly using multimedia and computing equipment.
- Diagnosed and resolved hardware and software issues, improving equipment reliability and operational efficiency.

PROJECTS

FindMyParking | UCI COMPSCI 125 (Next Generation Search Systems)

Jan. 2026 - Mar. 2026

- Developed a personalized parking recommendation engine that ranks parking spots based on cost, distance, availability, and user preferences.
- Built a destination-based search feature and implemented a multi-factor scoring algorithm combining parking cost, walking distance, traffic conditions, and user preferences to optimize a context-aware ranking algorithm
- Integrated real-time context signals including weather data (OpenWeatherMap) and traffic conditions (Mapbox Directions API) to dynamically adjust recommendations.
- Implemented geospatial queries using PostgreSQL + PostGIS to calculate proximity between parking locations, user position, and destination coordinates.
- Developed an interactive map visualization using Streamlit and PyDeck to display recommended parking locations relative to the user and destination.

SPENCER | Accessibility Product Design Project

Sep. 2025

- Designed a location-based mobile application prototype enabling users to locate hygiene product dispensers in public spaces.
- Developed user flows and interactive prototypes supporting location discovery and product availability tracking.
- Applied UX engineering principles to design responsive interface interactions and improve accessibility of location-based services.

Interlude | Personal Project - Spotify Recommendation App

Jul. 2025

- Developed a music recommendation platform that analyzes user listening histories via the Spotify Web API to surface mutually undiscovered tracks.
- Implemented algorithms to compare listening patterns between users, identifying overlapping artists, genres, and listening frequencies to generate recommendations.
- Built backend services in Node.js to handle API data retrieval, processing, and recommendation scoring.
- Designed a filtering and ranking pipeline to prioritize tracks based on similarity metrics and listening behavior.